

16/09/26

Sayab

Assignment

cloud Architecture

Chapter

Ans-2

cloud computing has the following major characteristics

1. On-demand self-service: user can automatically provision computing resources whenever required without human interaction with the provider.
2. Broad network access: cloud services can be accessed through the internet using computers, mobiles, and other devices.
3. Resource pooling: cloud providers share computing resources such as CPU, memory, and storage among multiple users.
4. Rapid elasticity: Resources can be quickly increased or decreased according to workload.
5. Measured service: cloud usage is monitored and users generally pay according to their resource consumption.

Ans-2

cloud

* Public cloud → cloud infrastructure is provided by a third-party provider and shared among multiple customers.

* Private cloud → cloud infrastructure is dedicated to one organization.

* Hybrid cloud → combination of private and public cloud clouds that can work together.

* Community cloud → Infrastructure is shared by organizations having common requirements or objectives

Suitable model for a university Hybrid cloud :-

A hybrid cloud is suitable for a university because.

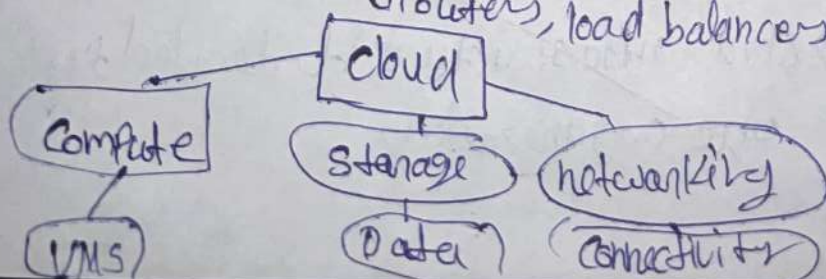
- Sensitive student and examination data can be kept in a Private cloud.
- Public websites and learning applications can use the Public cloud
- During examinations, additional public cloud resources can be used to handle increased traffic
- It provides flexibility, scalability and better resource utilization

Q.3

1. Compute :- Compute provides processing power to run applications and services. Examples include virtual machines and cloud servers.

2. Storage :- Storage is used to store files, databases, backups and application data, it can provide scalable storage according to requirements.

3. Networking :- Networking connects cloud resources and users. It includes virtual network, IP address, routers, load balancers and firewalls.



QAM=H
X=

SLA (Service Level Agreement) is an agreement between a cloud service provider and its customers that defines the expected level of service.

Importance :-

- Defines responsibility of the provider and customer
- Specifies expected service quality
- Helps measure provider performance
- Defines actions or compensation when agreed service levels are not met.

Two important SLA components

1. Availability/Uptime → Specifies the expected availability of the service, such as 99.9% uptime
2. Performance → Defines expected response time, latency or other performance requirements

Ans=5
X=

Suitable OpenStack components are

* Requirement

- Authentication
- VM Provisioning
- Networking
- Storage

OpenStack component

Keystone

Nova

Neutron

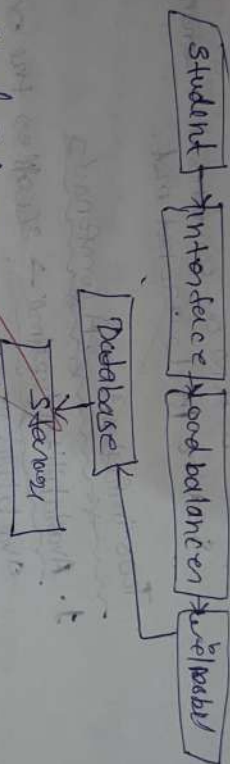
Cinder/Swift

- Keystone → Identity and authentication
- Nova → Compute and virtual machine provisioning
- Neutron → Networking
- Cinder → Block storage
- Swift → Object storage

Ques 6 → A university can design an online examination platform using distributed cloud architecture concept.

1. Cloud Computing Architecture :-

- Frontend — student and teacher web/mobile interface
- Application layer → Examination application
- Database → student question and results
- Cloud infrastructure → computer, storage and networking



2. Conceptual Reference Model :-

- SaaS → online examination and student data
- PaaS → application development/testing platform
- IaaS → virtual machines, networking and storage

3. Service Deployment :-

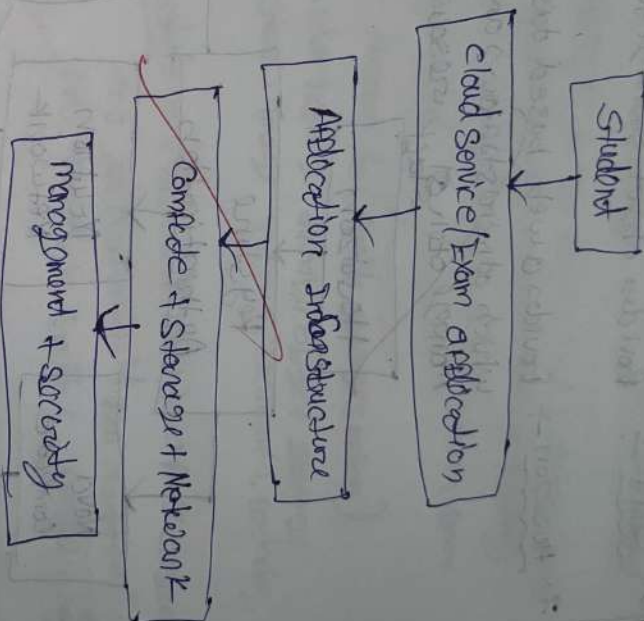
- Private cloud → sensitive examination and student data
- Public cloud → web application and administrative

4. Cloud Service Management :-

- Resource provisioning
- Monitoring
- Security
- Performance
- Billing/Resource

5. IBM CERA

IBM cloud ~~was~~ ^{comprising} Red Hat OpenShift Container Engine (CORA) provides a structured framework for designing cloud solutions. It helps identity critical services, infrastructure, management, security and overall a self-governing stateful cloud.



Ans: 7

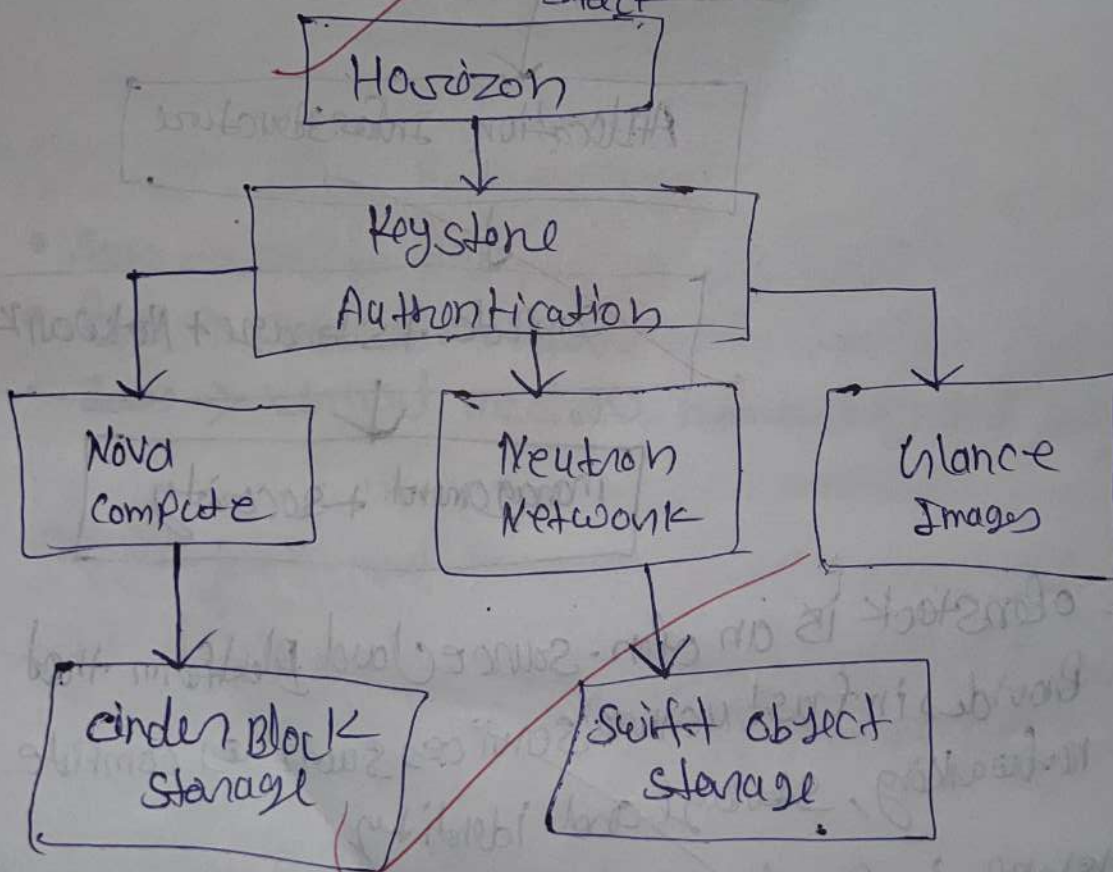
OpenStack is an open-source cloud platform that provides infrastructure services such as compute, networking, storage and identity.

KeyStone :- Provide identity and authentication.

2. Neutron :- Provide compute service and manages virtual machines.

3. Neutron :- Provide networking services and manages networks, subnets, routers and security groups.

4. Glance → Provide VM images used to create virtual machines.
5. Cinder → Provide persistent block storage for virtual machines.
6. Swift → Provides object storage for files and objects.
7. Horizon → Provides a web-based dashboard through which administrators and users can manage OpenStack resources.

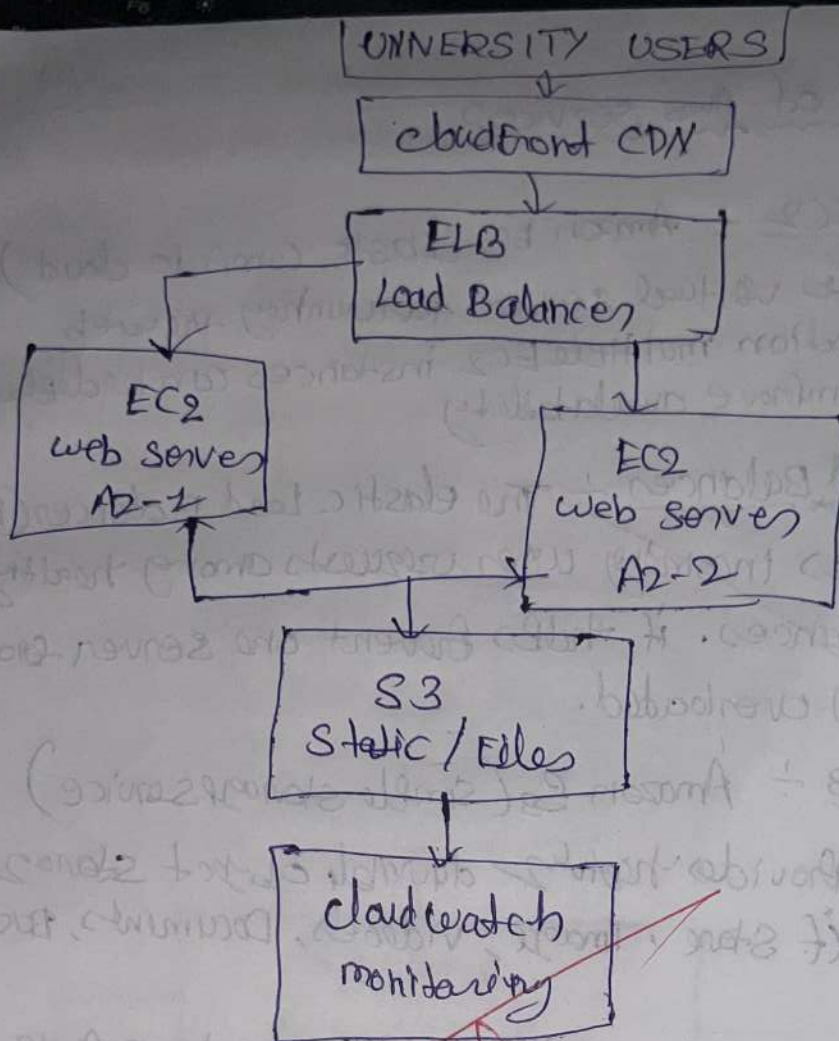


- user logs into Horizon.
- Keystone authenticates the user.
- user requests a new VM.
- Nova handles the compute request.
- Glance creates / connects the required network.

Ans = 8

Role of AWS services

1. Amazon EC2 ÷ Amazon EC2 (Elastic compute cloud)
Provides virtual servers for running the web application. Multiple EC2 instances can be deployed to improve availability.
2. Elastic Load Balances ÷ The elastic load balancer (ELB) distributes incoming user requests among healthy EC2 instances. It helps prevent one server from becoming overloaded.
3. Amazon S3 ÷ Amazon S3 (Simple storage service)
Provides highly durable object storage for store images, videos, documents, backups.
4. Amazon CloudWatch ÷ CloudWatch monitors AWS resources and applications. It can monitor CPU, Network traffic, applications, ^{alarms} logs.
5. Amazon CloudFront ÷ CloudFront is a content delivery network (CDN) that caches content at edge locations closer to users, reducing latency and decreasing the load on the origin.



* Working

- Users access the university web application
- CloudFront serves cached content from nearby edge locations
- Requests requiring the origin go toward the application through the ELB
- ELB distributes requests among healthy EC2 instances.
- Application files/static objects can be stored in S3
- Deploying instances across separate availability zones improves resilience against an individual zone failure